

## Lecture 6 (9/25)

*Variance, Normal models and Linear Regression*

# Variance and standard deviation

In addition to the expectation, we want to measure how spread out a distribution is. For this we define the variance

$$\text{var}(Y) = E[(Y - E[Y])^2]$$

# Variance and standard deviation

In addition to the expectation, we want to measure how spread out a distribution is. For this we define the variance

$$\text{var}(Y) = E[(Y - E[Y])^2]$$

Using

$$(Y - E[Y])^2 = Y^2 - 2YE[Y] + E[Y]^2$$

# Variance and standard deviation

In addition to the expectation, we want to measure how spread out a distribution is. For this we define the variance

$$\text{var}(Y) = E[(Y - E[Y])^2]$$

Using

$$(Y - E[Y])^2 = Y^2 - 2YE[Y] + E[Y]^2$$

and that expectation is a linear operation

$$\text{var}(Y) = E[Y^2] - 2E[Y]^2 + E[Y]^2 = E[Y^2] - E[Y]^2$$

# Variance and standard deviation

In addition to the expectation, we want to measure how spread out a distribution is. For this we define the variance

$$\text{var}(Y) = E[(Y - E[Y])^2]$$

Using

$$(Y - E[Y])^2 = Y^2 - 2YE[Y] + E[Y]^2$$

and that expectation is a linear operation

$$\text{var}(Y) = E[Y^2] - 2E[Y]^2 + E[Y]^2 = E[Y^2] - E[Y]^2$$

In code `np.var(y)`.

# Variance and standard deviation

In addition to the expectation, we want to measure how spread out a distribution is. For this we define the variance

$$\text{var}(Y) = E[(Y - E[Y])^2]$$

Using

$$(Y - E[Y])^2 = Y^2 - 2YE[Y] + E[Y]^2$$

and that expectation is a linear operation

$$\text{var}(Y) = E[Y^2] - 2E[Y]^2 + E[Y]^2 = E[Y^2] - E[Y]^2$$

In code `np.var(y)`. **Properties:**

- **Linear transformations:** For constants  $a$  and  $b$ ,  $\text{var}(aY + b) = a^2 \text{var}(Y)$ .

# Variance and standard deviation

In addition to the expectation, we want to measure how spread out a distribution is. For this we define the variance

$$\text{var}(Y) = E[(Y - E[Y])^2]$$

Using

$$(Y - E[Y])^2 = Y^2 - 2YE[Y] + E[Y]^2$$

and that expectation is a linear operation

$$\text{var}(Y) = E[Y^2] - 2E[Y]^2 + E[Y]^2 = E[Y^2] - E[Y]^2$$

In code `np.var(y)`. **Properties:**

- **Linear transformations:** For constants  $a$  and  $b$ ,  $\text{var}(aY + b) = a^2 \text{var}(Y)$ .
- **Sums of independent variables:** If  $X$  and  $Y$  are independent,  $\text{var}(X + Y) = \text{var}(X) + \text{var}(Y)$ .

# Variance and standard deviation

In addition to the expectation, we want to measure how spread out a distribution is. For this we define the variance

$$\text{var}(Y) = E[(Y - E[Y])^2]$$

Using

$$(Y - E[Y])^2 = Y^2 - 2YE[Y] + E[Y]^2$$

and that expectation is a linear operation

$$\text{var}(Y) = E[Y^2] - 2E[Y]^2 + E[Y]^2 = E[Y^2] - E[Y]^2$$

In code `np.var(y)`. **Properties:**

- **Linear transformations:** For constants  $a$  and  $b$ ,  $\text{var}(aY + b) = a^2 \text{var}(Y)$ .
- **Sums of independent variables:** If  $X$  and  $Y$  are independent,  $\text{var}(X + Y) = \text{var}(X) + \text{var}(Y)$ .

**Related quantities:**

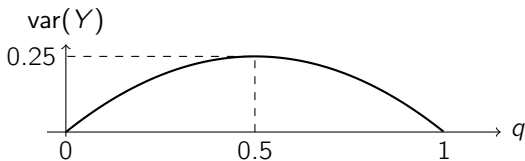
- The standard deviation is  $\sigma = \sqrt{\text{var}(Y)}$
- The coefficient of variation is  $\text{CV}_Y = \sigma/\mu$  where  $\mu = E[Y]$ .

## Example

Let  $Y \sim \text{Bernoulli}(q)$ . Compute  $\text{var}(Y)$ .

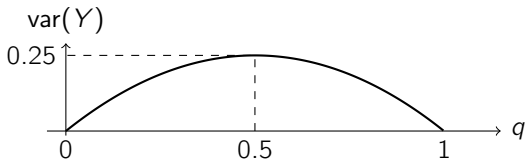
## Example

Let  $Y \sim \text{Bernoulli}(q)$ . Compute  $\text{var}(Y)$ .



### Example

Let  $Y \sim \text{Bernoulli}(q)$ . Compute  $\text{var}(Y)$ .



### Example

Let  $Y \sim \text{Uniform}(0, L)$ . Compute  $\text{var}(Y)$  and CV.

# The Normal distribution

$Y \sim \text{Normal}(\mu, \sigma^2)$  has density

$$g(y) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(y-\mu)^2/2\sigma^2},$$

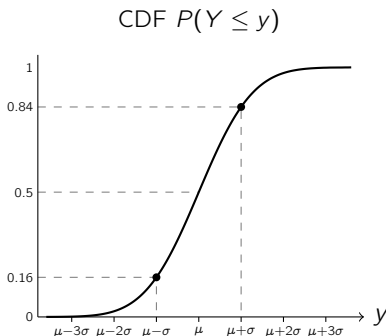
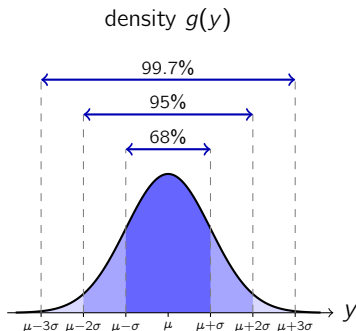
with  $E[Y] = \mu$  and  $\text{var}(Y) = \sigma^2$ .

# The Normal distribution

$Y \sim \text{Normal}(\mu, \sigma^2)$  has density

$$g(y) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(y-\mu)^2/2\sigma^2},$$

with  $E[Y] = \mu$  and  $\text{var}(Y) = \sigma^2$ .



# Properties

- The Normal distribution is the “most random”<sup>1</sup> random variable with a given mean and variance.

---

<sup>1</sup>In the sense of max entropy

# Properties

- The Normal distribution is the “most random”<sup>1</sup> random variable with a given mean and variance.
- If  $Y \sim \text{Normal}(\mu, \sigma^2)$ , then

$$Z = \frac{Y - \mu}{\sigma} \sim \text{Normal}(0, 1), \quad aY + b \sim \text{Normal}(a\mu + b, a^2\sigma^2).$$

---

<sup>1</sup>In the sense of max entropy

# Properties

- The Normal distribution is the “most random”<sup>1</sup> random variable with a given mean and variance.
- If  $Y \sim \text{Normal}(\mu, \sigma^2)$ , then

$$Z = \frac{Y - \mu}{\sigma} \sim \text{Normal}(0, 1), \quad aY + b \sim \text{Normal}(a\mu + b, a^2\sigma^2).$$

- If  $X \sim \text{Normal}(\mu_X, \sigma_X^2)$  and  $Y \sim \text{Normal}(\mu_Y, \sigma_Y^2)$  are independent, then

$$X + Y \sim \text{Normal}(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2)$$

No other distribution with finite mean and variance has these properties!

---

<sup>1</sup>In the sense of max entropy

# Properties

- The Normal distribution is the “most random”<sup>1</sup> random variable with a given mean and variance.
- If  $Y \sim \text{Normal}(\mu, \sigma^2)$ , then

$$Z = \frac{Y - \mu}{\sigma} \sim \text{Normal}(0, 1), \quad aY + b \sim \text{Normal}(a\mu + b, a^2\sigma^2).$$

- If  $X \sim \text{Normal}(\mu_X, \sigma_X^2)$  and  $Y \sim \text{Normal}(\mu_Y, \sigma_Y^2)$  are independent, then

$$X + Y \sim \text{Normal}(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2)$$

No other distribution with finite mean and variance has these properties!

---

<sup>1</sup>In the sense of max entropy

# Conditional Normal Model

A (single-predictor) linear regression model is conditionally Normal:

$$Y \mid X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2).$$

# Conditional Normal Model

A (single-predictor) linear regression model is conditionally Normal:

$$Y \mid X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2).$$

Equivalently, when  $X = x$ ,

$$Y = \beta_0 + \beta_1 x + \epsilon, \quad \epsilon \sim \text{Normal}(0, \sigma_\epsilon^2).$$

# Conditional Normal Model

A (single-predictor) linear regression model is conditionally Normal:

$$Y \mid X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2).$$

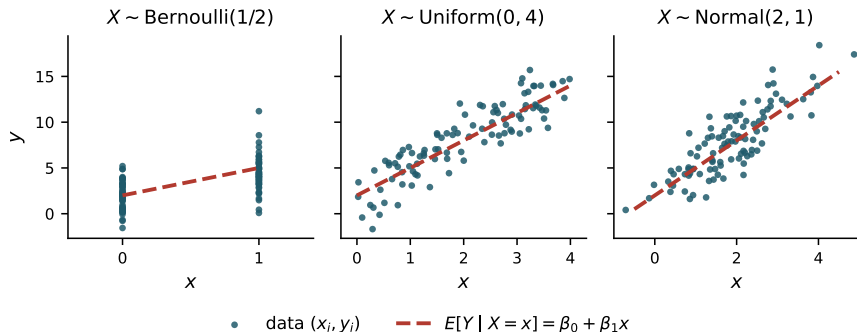
Equivalently, when  $X = x$ ,

$$Y = \beta_0 + \beta_1 x + \epsilon, \quad \epsilon \sim \text{Normal}(0, \sigma_\epsilon^2).$$

- $X$  is the predictor
- $Y$  is the response variable
- $\beta_1$  is called the regression coefficient of  $X$ .
- The variable  $\epsilon$  may be referred to as an exogenous noise source. Exogeneity means this noise is independent of  $X$ .

# What data from this model looks like

Data simulated from  $Y | X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\varepsilon^2)$



# Linear regression assumptions

In a linear regression model we make 3 assumptions:

# Linear regression assumptions

In a linear regression model we make 3 assumptions:

- 1) Conditional normality:  $Y \mid X$  is Normal.

# Linear regression assumptions

In a linear regression model we make 3 assumptions:

- 1) Conditional normality:  $Y \mid X$  is Normal.
- 2) Linearity of the conditional mean:  $E[Y \mid X = x] = \beta_0 + \beta_1 x$ .

# Linear regression assumptions

In a linear regression model we make 3 assumptions:

- 1) Conditional normality:  $Y | X$  is Normal.
- 2) Linearity of the conditional mean:  $E[Y | X = x] = \beta_0 + \beta_1 x$ . This means increasing  $x$  by one changes  $Y$  by  $\beta_1$ , on average. For any  $x$

$$\beta_1 = E[Y | X = x + 1] - E[Y | X = x]$$

# Linear regression assumptions

In a linear regression model we make 3 assumptions:

- 1) Conditional normality:  $Y | X$  is Normal.
- 2) Linearity of the conditional mean:  $E[Y | X = x] = \beta_0 + \beta_1 x$ . This means increasing  $x$  by one changes  $Y$  by  $\beta_1$ , on average. For any  $x$

$$\beta_1 = E[Y | X = x + 1] - E[Y | X = x]$$

- 3) Homoskedasticity:  $\text{var}(Y | X = x)$  does not depend on  $x$ .

# Marginals and covariance

Now suppose  $E[X] = \mu_X$  and  $\text{var}(X) = \sigma_X^2$ .  $X$  can be any distribution with this mean and variance.

# Marginals and covariance

Now suppose  $E[X] = \mu_X$  and  $\text{var}(X) = \sigma_X^2$ .  $X$  can be any distribution with this mean and variance. Calculate the following:

- $\mu_Y = E[Y]$ .
- $E[XY]$

# Marginals and covariance

Now suppose  $E[X] = \mu_X$  and  $\text{var}(X) = \sigma_X^2$ .  $X$  can be any distribution with this mean and variance. Calculate the following:

- $\mu_Y = E[Y]$ .
- $E[XY]$

The covariance is

$$\text{cov}(X, Y) = E[XY] - E[Y]E[X]$$

# Marginals and covariance

Now suppose  $E[X] = \mu_X$  and  $\text{var}(X) = \sigma_X^2$ .  $X$  can be any distribution with this mean and variance. Calculate the following:

- $\mu_Y = E[Y]$ .
- $E[XY]$

The covariance is

$$\text{cov}(X, Y) = E[XY] - E[Y]E[X]$$

and from the calculations above

$$\beta_1 = \frac{\text{cov}(X, Y)}{\sigma_X^2}$$

# Marginals and covariance

Now suppose  $E[X] = \mu_X$  and  $\text{var}(X) = \sigma_X^2$ .  $X$  can be any distribution with this mean and variance. Calculate the following:

- $\mu_Y = E[Y]$ .
- $E[XY]$

The covariance is

$$\text{cov}(X, Y) = E[XY] - E[Y]E[X]$$

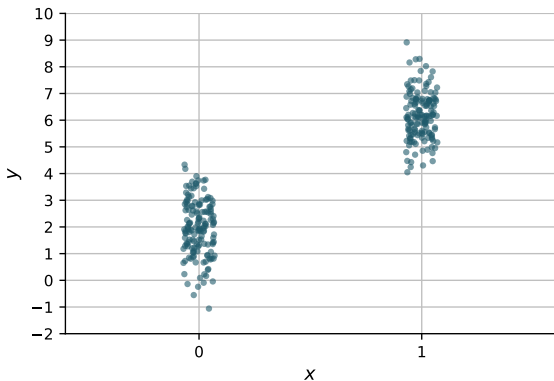
and from the calculations above

$$\boxed{\beta_1 = \frac{\text{cov}(X, Y)}{\sigma_X^2}}$$

This formula connects the regression coefficient to the covariance (and hence to expectations we can compute from data).

## Example

Write down a linear regression model for the following data, where you can assume  $X \sim \text{Bernoulli}(1/2)$ . What is  $\text{cov}(X, Y)$  and  $\text{var}(Y)$ ?



## Example

Two attributes  $X$  and  $Y$  (e.g. income and height) have a covariance of 2. An increase of  $X$  by two units leads to an increase in  $Y$  of 1, on average. Assuming a linear regression model  $Y | X$ , what is the variance of  $X$ ?

# Marginal variance

Calculate  $\sigma_Y^2 = \text{var}(Y)$ , where  $Y \mid X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2)$

$$\text{var}(Y) = \text{var}(\beta_0 + \beta_1 X + \epsilon)$$

# Marginal variance

Calculate  $\sigma_Y^2 = \text{var}(Y)$ , where  $Y | X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2)$

$$\begin{aligned}\text{var}(Y) &= \text{var}(\beta_0 + \beta_1 X + \epsilon) \\ &= \text{var}(\beta_1 X + \epsilon) \quad (\text{adding a constant doesn't change the variance})\end{aligned}$$

# Marginal variance

Calculate  $\sigma_Y^2 = \text{var}(Y)$ , where  $Y | X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2)$

$$\begin{aligned}\text{var}(Y) &= \text{var}(\beta_0 + \beta_1 X + \epsilon) \\ &= \text{var}(\beta_1 X + \epsilon) && \text{(adding a constant doesn't change the variance)} \\ &= \text{var}(\beta_1 X) + \text{var}(\epsilon) && \text{(variances add for independent variables)}\end{aligned}$$

# Marginal variance

Calculate  $\sigma_Y^2 = \text{var}(Y)$ , where  $Y \mid X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2)$

$$\begin{aligned}\text{var}(Y) &= \text{var}(\beta_0 + \beta_1 X + \epsilon) \\ &= \text{var}(\beta_1 X + \epsilon) && \text{(adding a constant doesn't change the variance)} \\ &= \text{var}(\beta_1 X) + \text{var}(\epsilon) && \text{(variances add for independent variables)} \\ &= \beta_1^2 \text{var}(X) + \text{var}(\epsilon) && (\text{var}(aX) = a^2)\end{aligned}$$

# Marginal variance

Calculate  $\sigma_Y^2 = \text{var}(Y)$ , where  $Y \mid X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2)$

$$\begin{aligned}\text{var}(Y) &= \text{var}(\beta_0 + \beta_1 X + \epsilon) \\&= \text{var}(\beta_1 X + \epsilon) && \text{(adding a constant doesn't change the variance)} \\&= \text{var}(\beta_1 X) + \text{var}(\epsilon) && \text{(variances add for independent variables)} \\&= \beta_1^2 \text{var}(X) + \text{var}(\epsilon) && (\text{var}(aX) = a^2) \\&= \beta_1^2 \sigma_X^2 + \sigma_\epsilon^2 && \text{(definitions of } \sigma_X^2 \text{ and } \sigma_\epsilon^2)\end{aligned}$$

# Standardized variables

Let  $Y | X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2)$  be two random variables related by a linear regression model.

# Standardized variables

Let  $Y | X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2)$  be two random variables related by a linear regression model.

Consider the standardized variables

$$Z_X = (X - \mu_X)/\sigma_X, \quad Z_Y = (Y - \mu_Y)/\sigma_Y$$

What is the regression coefficient when regressing  $Z_Y$  on  $Z_X$ ?

# Standardized variables

$$Z_Y = (Y - \mu_Y) / \sigma_Y$$

## Standardized variables

$$\begin{aligned} Z_Y &= (Y - \mu_Y) / \sigma_Y \\ &= [(\beta_0 + \beta_1 X + \epsilon) - (\beta_0 + \beta_1 \mu_X)] / \sigma_Y \quad (\text{substitute } Y, \mu_Y) \end{aligned}$$

# Standardized variables

$$\begin{aligned}Z_Y &= (Y - \mu_Y)/\sigma_Y \\&= [(\beta_0 + \beta_1 X + \epsilon) - (\beta_0 + \beta_1 \mu_X)]/\sigma_Y \quad (\text{substitute } Y, \mu_Y) \\&= [\cancel{\beta_0} + \beta_1 X + \epsilon - \cancel{\beta_0} - \beta_1 \mu_X]/\sigma_Y \quad (\beta_0 \text{ cancels})\end{aligned}$$

# Standardized variables

$$\begin{aligned}Z_Y &= (Y - \mu_Y)/\sigma_Y \\&= [(\beta_0 + \beta_1 X + \epsilon) - (\beta_0 + \beta_1 \mu_X)]/\sigma_Y && \text{(substitute } Y, \mu_Y) \\&= [\cancel{\beta_0} + \beta_1 X + \epsilon - \cancel{\beta_0} - \beta_1 \mu_X]/\sigma_Y && (\beta_0 \text{ cancels}) \\&= [\beta_1(X - \mu_X) + \epsilon]/\sigma_Y && \text{(factor out } \beta_1)\end{aligned}$$

# Standardized variables

$$\begin{aligned}Z_Y &= (Y - \mu_Y)/\sigma_Y \\&= [(\beta_0 + \beta_1 X + \epsilon) - (\beta_0 + \beta_1 \mu_X)]/\sigma_Y && \text{(substitute } Y, \mu_Y) \\&= [\cancel{\beta_0} + \beta_1 X + \epsilon - \cancel{\beta_0} - \beta_1 \mu_X]/\sigma_Y && (\beta_0 \text{ cancels}) \\&= [\beta_1(X - \mu_X) + \epsilon]/\sigma_Y && \text{(factor out } \beta_1) \\&= \frac{\beta_1 \sigma_X}{\sigma_Y} \cdot \frac{X - \mu_X}{\sigma_X} + \frac{\epsilon}{\sigma_Y} && \text{(multiply, divide by } \sigma_X)\end{aligned}$$

# Standardized variables

$$\begin{aligned}Z_Y &= (Y - \mu_Y)/\sigma_Y \\&= [(\beta_0 + \beta_1 X + \epsilon) - (\beta_0 + \beta_1 \mu_X)]/\sigma_Y && \text{(substitute } Y, \mu_Y) \\&= [\cancel{\beta_0} + \beta_1 X + \epsilon - \cancel{\beta_0} - \beta_1 \mu_X]/\sigma_Y && (\beta_0 \text{ cancels}) \\&= [\beta_1(X - \mu_X) + \epsilon]/\sigma_Y && \text{(factor out } \beta_1) \\&= \frac{\beta_1 \sigma_X}{\sigma_Y} \cdot \frac{X - \mu_X}{\sigma_X} + \frac{\epsilon}{\sigma_Y} && \text{(multiply, divide by } \sigma_X) \\&= \frac{\beta_1 \sigma_X}{\sigma_Y} Z_X + \frac{\epsilon}{\sigma_Y} && (Z_X = (X - \mu_X)/\sigma_X)\end{aligned}$$

# Standardized variables

$$\begin{aligned}Z_Y &= (Y - \mu_Y)/\sigma_Y \\&= [(\beta_0 + \beta_1 X + \epsilon) - (\beta_0 + \beta_1 \mu_X)]/\sigma_Y && \text{(substitute } Y, \mu_Y) \\&= [\cancel{\beta_0} + \beta_1 X + \epsilon - \cancel{\beta_0} - \beta_1 \mu_X]/\sigma_Y && (\beta_0 \text{ cancels}) \\&= [\beta_1(X - \mu_X) + \epsilon]/\sigma_Y && \text{(factor out } \beta_1) \\&= \frac{\beta_1 \sigma_X}{\sigma_Y} \cdot \frac{X - \mu_X}{\sigma_X} + \frac{\epsilon}{\sigma_Y} && \text{(multiply, divide by } \sigma_X) \\&= \frac{\beta_1 \sigma_X}{\sigma_Y} Z_X + \frac{\epsilon}{\sigma_Y} && (Z_X = (X - \mu_X)/\sigma_X)\end{aligned}$$

Therefore

$$\boxed{\beta_1 \sigma_X / \sigma_Y = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}}$$

This is called the correlation coefficient and denoted  $\rho$ .

# Solutions

- 1)  $Y \sim \text{Bernoulli}(q)$ :  $\text{var}(Y) = q(1 - q)$ .
- 2)  $Y \sim \text{Uniform}(0, L)$ :  $\text{var}(Y) = \frac{L^2}{12}$ ,  $\text{CV} = \frac{L/\sqrt{12}}{L/2} = \frac{1}{\sqrt{3}} \approx 0.58$ .
- 3)  $\mu_Y = \beta_0 + \beta_1 \mu_X$  and  $E[XY] = \beta_0 \mu_X + \beta_1(\sigma_X^2 + \mu_X^2)$ , so  $\text{cov}(X, Y) = \beta_1 \sigma_X^2$ .
- 4) The group means are about  $\approx 2$  and  $6$ , so  $\beta_1 \approx 4$ .  $\sigma_X^2 = \frac{1}{4}$  and  $\sigma_\epsilon \approx 1$ . Hence  $\text{cov}(X, Y) \approx \beta_1 \sigma_X^2 = 1$  and  $\text{var}(Y) \approx \beta_1^2 \sigma_X^2 + \sigma_\epsilon^2 = 5$ .
- 5)  $\beta_1 = \frac{1}{2}$ , so  $\sigma_X^2 = \text{cov}(X, Y)/\beta_1 = 4$ .